3D Reconstruction of Cerebral Blood Vessels from MRI via Frangi Vesselness Filtering

Thiago Rodrigues, Yonathan Tesfaye

Abstract

Reconstructing cerebral blood vessels from MRI scans is challenging due to its noise, low contrast, and difficulty in detecting thin vascular structures. This project explores the use of the Frangi Vesselness Filter as a traditional, interpretable technique used for improving tubular features in 2D MRI slices and constructing a 3D vascular model. Applying the filter, marching cubes reconstruction, and threshold segmentation, we are capable of obtaining a reliable 3D approximation of major brain regions, without the need for any type of deep learning or radiation based imaging. While small vessels remain difficult to recover, the results show that this pipeline produces an accessible and efficient approach for MRI based vascular reconstruction.

I. Introduction

Throughout the years, research in medical imaging had made a significant amount of advancements. Especially when taking into consideration the analysis and visualization of vascular structures. The precision and accuracy in the reconstruction of blood vessels is currently crucial in many areas of medical imaging. Some of these areas are diagnosis, surgical planning, and treatment of cardiovascular diseases. Even after all of this, accurately reconstructing 3D vessel structures from a 2D medical plane scans with minor errors has proven to be quite the challenge.

The main goal of our research project is to design a Frangi vessel filter capable if reconstructing a 3D vascular structure from a 2D medical scan. In medicine, the most common used methodology is Computer Tomography Angiography (CTA). Computed Tomography Angiography involves a combination between CT scans with a injection of a special dye agent into blood vessels and tissues which, highlights blood vessels for clearer images. This method allows accurate mapping and imaging of the blood vessels to prevent any health concerns.

The only problem with this is that by allowing CTA's, patients will have to be exposed to invasive radiation that could potentially harm them. That is because, doctors take countless X-Ray scans of the brain, meaning the patient will be exposed to radiation several times throughout their treatment. Then after acquiring medical images of the brain, they use reconstruction algorithms. Projection, surface display, volume rendering and much more, just to build a 3D model of the brain blood vessels.

This research proposes a fundamentally different algorithm that limits the amount of foreign radiation into patients bodies, and have a better prediction towards the reconstruction of the brains. Traditional methods rely heavily on rule based algorithms, manual segmentation, or geometric modeling, all of which can be often considered as labor intensive, costly and time consuming. these approaches can also introduce human error and raise issues about reliability and consistency. Looking into the future, by moving beyond these traditional techniques, doctors and researcher are capable of achieving greater efficient and accuracy in the 3D reconstruction of brain vessels. This shift will also improve workflow and open the door for future advancements in medical analysis and imaging.

Looking into vessel reconstruction, which relays heavily on traditional techniques. All of these approaches provide imperfect solutions, they often struggle with generalization, specifically when confronted with varying imaging conditions and unique information on patients. On the other hand, using the Frangi vessel filter has the capability to learn complex vessel features. It is capable of assimilating spatial connections directly from large datasets. This means it would increase accuracy while reducing human errors.

Taking this into consideration, the purpose of our study is to do an in depth exploration of the Frangi Vesselness Filter as a key method for changing two-dimensional medical images into three-dimensional reconstructions of brain blood vessels. Rather than relying on deep learning methods, this research had a greater focus on optimizing and understanding the multiscale, Hessian principles underlying the Frangi Vesselness Filter. In greater detail, this project's goal is to analyze how the filter enhances tubular structures, and how 2D slices can be systematically fused into a coherent volumetric 3D model. By applying the Frangi vesselness filter to each cross-sectional image, we seek to improve the continuity, clarity and detectability of vascular structures prior to reconstruction. Through this focused investigation, we also aim to highlight the potential of the Frangi vesselness filter as an interpretable and highly efficient technique for reconstructing and enhancing cerebral blood vessels in medical imaging.

A. Related Methods

A wide range of techniques has been developed to analyze and extract vascular structures in medical imaging. When thinking about medical imaging, we particularly think about angiography and MRI. Traditional segmentation approaches often relied on techniques such as model based tracking, region growing, and edge detection. While these methods provide the basic foundation for vessel extraction, they are often highly sensitive to noise, complex vascular morphology, and intensity variations, which limit their reliability in clinical environments.

Recently, deep learning has become a main topic in vascular reconstruction and segmentation. Models such as AngioNet, specifically designed for angiography vessel extraction and domain aware architectures to detect low contrast and thin vessels more reliably, rather than traditional rule based methods.

On the other hand, CNNs form the base of many modern pipelines. U-Net established a powerful framework for 2D medical segmentation by utilizing skip connections to save spatial detail, while 3D U-Net and V-Net extend these ideologies to volumetric data through full 3D convolution to address severe class imbalance.

However, these deep learning models achieve a promising performance. They require large annotated datasets, careful tuning, and especially high computational cost. These constraints emphasize the need for mathematically grounded alternatives, such as the method explored within this study.

B. Problem formulation

Reconstructing the brain vessels from MRI data remains challenging, since MRI often provides a limited contrast between surrounding tissue and vascular structures. Low intensity and small vessels are easily concealed by noise, partial volume effects, and intensity variations, making them difficult to separate using traditional segmentation methods. These limitations often lead to structural discontinuities and fragmented vessel maps, which compromise the accuracy of any 3D reconstruction.

The main problem within this study is how to extract and enhance tubular vascular structure from 2D MRI slices in a way that preserves anatomical integrity and continuity. To achieve this, it requires a filtering method capable of amplifying patterns that look like a vessel while suppressing nonvascular features and noise. Doing all of this without relying on deep learning or large annotated datasets. The central goal is to determine how effectively the Frangi Vesselness Filter can improve the visibility of brain vessels and produce a consistent output.

II. Methodology

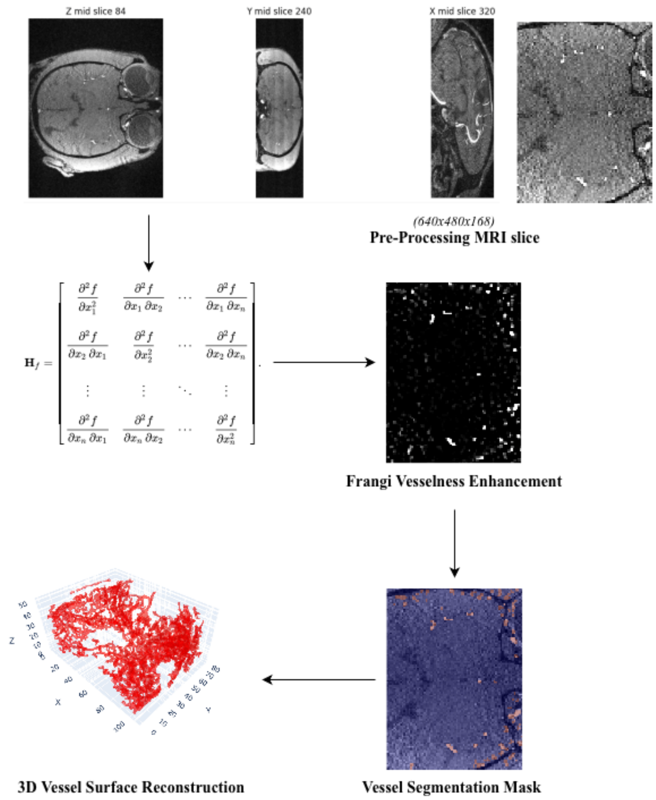

Fig. 1. Pipeline for reconstructing 3D brain vasculature from MRI slices. (A) A representative slice from the input MRI volume. (B) Vessel-enhanced image produced using the multiscale Frangi filter, highlighting tubular structures. (C) Binary vessel segmentation obtained through thresholding. (D) 3D surface reconstruction of the segmented vasculature using the marching cubes algorithm.

In this research we will be applying a Frangi filter to reconstruct 2D volumetric slides. We propose an automated image-processing pipeline for vessel segmentation and 3D model generation. The approach utilizes a known "vesselness" filter (to enhance vessels in the images) followed by threshold-based segmentation and volumetric rendering. This method is applicable to implement in Python (with libraries like scikit-image, NiBabel, etc.) and avoids tedious software or lengthy neural network training. See Fig.1 for general method overview.

A. Data Acquisition

The MRI data used in this project were obtained from the OpenNeuro public neuroimaging repository. The dataset includes a Time-of-Flight Magnetic Resonance Angiography (TOF-MRA) scan. Which was composed of 8 TOF-MRA scans, of which 4 are scanned with 3 Tesla Siemens machine and other 4 are scanned with 7 Tesla Siemens machine (Cui Yue, et al. "FFCM-MRF: An accurate and generalizable cerebrovascular segmentation pipeline on TOF-MRA). After downloading the volume from the OpenNeuro subject page, the file was imported into our processing pipeline using the NiBabel library in Python. The data consist of 2D axial slices, which we treated as the basis for vessel enhancement and reconstruction. The volumes were downsized due to enormous wait time while running the Frangi Fitler. No additional preprocessing steps, such as denoising or skull stripping were applied, as the objective was to evaluate whether classical image-processing techniques could reconstruct from the minimally preprocessed OpenNeuro MRI volume.

B. Preprocessing the MRI Slices

We first load the 2D MRI slice series and arrange them into the correct order to form a 3D volume. Basic preprocessing can be done if needed – for example, applying a slight Gaussian smoothing to reduce noise or normalizing intensity across slices. (In our case, time-of-flight MRA images already highlight blood vessels against dark background, so extensive preprocessing was not necessary).

C. Vessel Enhancement Filtering

Next, we apply a Frangi vesselness filter to the volume. This filter is specifically designed to detect continuous tubular structures like blood vessels by analyzing the second-order image derivatives (the Hessian matrix) at multiple scales (researchgate.net). In essence, the filter brightens line-like features and suppresses blob-like or plate-like structures, making the thin vessels stand out from surrounding tissue. We use the open-source implementation available in scikit-image (which supports 2D/3D images) to perform this enhancement (researchgate.net). After this step, we obtain a vessel-enhanced image where voxels belonging to vessels have high intensity values. Note: The Frangi filter may require some parameter tuning (scale range, sensitivity parameters) to optimally capture very fine vessel, due to our limited resources and computational prowess this was not performed (sarxiv.org). Additionally, given our time constraint, we started with default parameters and a reasonable scale range and visually inspect if the small vessels appear

D. Threshold-Based Segmentation

With the vesselness-enhanced volume, we perform segmentation by thresholding. The idea is to convert the continuous vesselness response into a binary mask that labels each voxel as either "vessel" or "background." We chose an intensity threshold where vessel voxels (bright in the filtered image) are separated from noise/background. To preserve the continuity of thin vessels, we can use hysteresis thresholding (dual threshold) – e.g. choose an upper threshold to definitely include strong vessel signals and a lower threshold to include weaker connected voxels as long as they attach to high-confidence areas (arxiv.org). This two-tier threshold helps maintain connected vessel segments so that faint parts of a vessel aren't dropped if they connect to brighter segments (arxiv.org). The result of this step is a 3D binary vessel mask volume. We may apply minor morphological operations for cleanup: for example, remove tiny isolated spots not connected to the main vasculature (false positives), or fill small gaps if any. At this point, we have essentially a 3D reconstruction of the vessels in voxel form (each slice's vessels are delineated and stacked consistently). This classic vesselness+threshold method is a common baseline for vessel segmentation and has even been used to generate training labels in research (arxiv.org) – making it a reliable approach for our study.

E. 3D Model Generation

To present the vessels as a smooth 3D structure, we convert the binary volume mask into a surface mesh. We will use the Marching Cubes algorithm to extract an isosurface (3D surface) from the 3D mask at the vessel boundary (frontiersin.org). Marching Cubes is a standard method that creates a high-resolution 3D surface by triangulating the boundary between "inside" (vessel) and "outside" (background) voxels (Openai). Many open libraries support this. The output will be a triangulated mesh of the vascular tree. We can then visualize this 3D model interactively or in a rendered image for our presentation. The mesh shows the full geometry of cerebral arteries, which we can color or annotate if needed. This fulfills the goal of obtaining a full 3D model of the brain vessels from the original slices.

III. Results

The multiscale Frangi filter generated high vesselness responses along tubular structures within the MRI volume, despite the absence of preprocessing. Thresholding proved to be an accurate capture of the geometry of the larger cerebral vessels. In addition it successfully eliminated small false-positive regions. The 3D mesh reconstructed from the binary mask via marching cubes qualitatively revealed an anatomically consistent approximation of the vascular tree. Major vessel trunks were visible, although finer vascular detail was not recovered due to downsampling and the simplicity of the segmentation strategy. Fig.2 demonstrates that classical image-processing techniques can yield decent vascular reconstructions from OpenNeuro MRI data with minimal computational resources.

IV. Discussion

The results of this project show that the Frangi Vesselness Filter can successfully improve major brain vessels in MRI slices and enable a consistent 3D reconstruction using only traditional image processing techniques. Even with down sampled data and minimal pre processing, the vesselness responses were able to produce clear tubular improvements, and when looking at the threshold segmentation, which captured the overall geography of the major arteries. This emphasizes the fact that the Fragin vesselness filter remains an interpretable and reliable tool for vascular extraction in MRI. Although the reconstruction itself also showed the limitations of this technique. Fine vessels were often lost or incomplete because of the low MRI contrast and

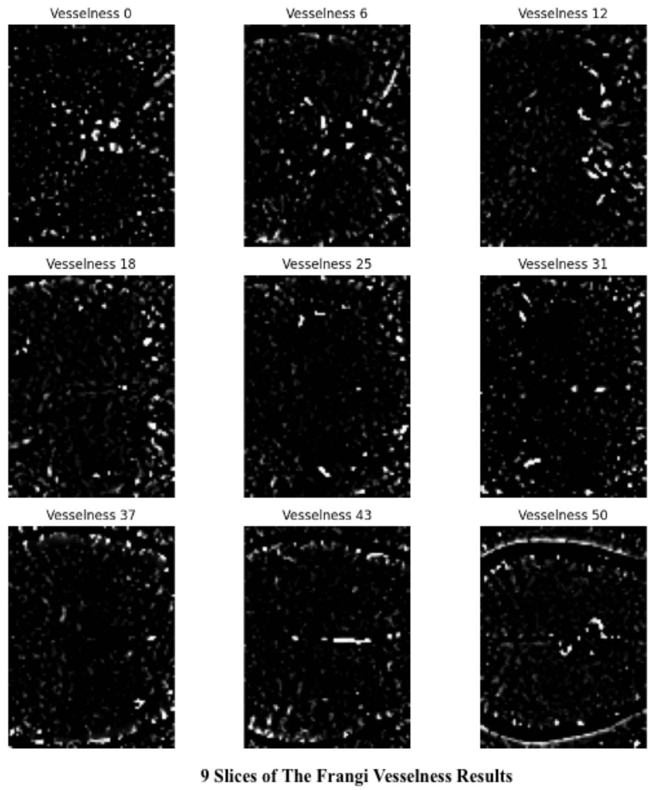

Fig. 2. Frangi vesselness responses across multiple slices, highlighting tubular vessel structures.

the simplicity of the threshold based segmentation. Also, when looking at the final mesh, it lacked the detail seen in higher resolution or deep learning based approaches. There were also computational constraints that hindered the scale range we could use, which harshly affected the visibility of smaller branches. However, despite the problems, the pipeline provided a promising foundation for traditional vessel reconstruction and emphasized the importance of the practical value of non deep learning methods in an environment with limited resources or data.

V. Conclusion

Finally, this project presented a traditional pipeline for reconstructing 3D brain vessels from MRI using the Frangi Vesselness Filter, marching cubes surface extraction, and threshold segmentation. The results show that this approach can successfully recover and improve the major vascular structures in the brain without relying on large datasets, deep learning, or radiation based imaging. However, the method definitely struggles to capture very small, precise vessels due to MRI's simple segmentation step and contrast limitations. Yet it remains efficient, practical, and interpretable for preliminary vascular analysis. In the future, this type of research should explore improved thresholding strategies, optimized filter parameters, and higher resolution data to increase the detail and continuity of the reconstructed vasculature.

Code Availability The code being used in this current research proposal is available here https://github.com/natifrr/Frangi-Filtering-on-Cerebral-Vessels

References

1. Frangi, A. F., Niessen, W. J., Vincken, K. L., and Viergever, M. A. (1998). Multiscale vessel enhancement filtering. In Medical Image Computing and Computer-Assisted Intervention — MICCAI 1998 (pp. 130–137). Springer.
2. Yue Cui, Haibin Huang, and Jialu Liu (2023). Macaque angiography dataset. OpenNeuro. [Dataset] https://doi.org/10.18112/openneuro.ds004620.v1.0.0
3. Lorensen, W. E., and Cline, H. E. (1987). Marching cubes: A high resolution 3D surface construction algorithm. ACM SIGGRAPH Computer Graphics, 21(4), 163–169.
4. Brett, M., Markiewicz, C. J., Hanke, M., et al. (2020). NiBabel: Accessing neuroimaging data formats in Python. Zenodo. https://doi.org/10.5281/zenodo.591597
5. van der Walt, S., Schönberger, J. L., Nunez-Iglesias, J., et al. (2014). scikit-image: Image processing in Python. PeerJ, 2:e453. https://doi.org/10.7717/peerj.453
6. Plotly Technologies Inc. (2015). Collaborative data science. https://plot.ly
7. Iyer, Kritika, et al. "AngioNet: A Convolutional Neural Network for Vessel Segmentation in X-Ray Angiography." Scientific Reports, vol. 11, no. 1, 10 Sept. 2021, p. 18066, www.nature.com/articles/s41598-021-97355-8, https://doi.org/10.1038/s41598-021-97355-8. Accessed 10 Aug. 2022.
8. Chapman, Brian E., and Dennis L. Parker. "3D Multi-Scale Vessel Enhancement Filtering Based on Curvature Measurements: Application to Time-of-Flight MRA." Medical Image Analysis, vol. 9, no. 3, June 2005, pp. 191–208, https://doi.org/10.1016/j.media.2004.08.001. Accessed 27 June 2020.
9. Si, W, and Y Feng. "A Multi-Channel Input Convolutional Neural Network for Artifact Reduction in Quantitative Susceptibility Mapping." Nan Fang Yi Ke Da Xue Xue Bao = Journal of Southern Medical University, vol. 42, no. 12, 2022, pp. 1799–1806, pubmed.ncbi.nlm.nih.gov/36651247/, https://doi.org/10.12122/j.issn.1673-4254.2022.12.07.
10.Aylward, S.R., and E. Bullitt. "Initialization, Noise, Singularities, and Scale in Height Ridge Traversal for Tubular Object Centerline Extraction." IEEE Transactions on Medical Imaging, vol. 21, no. 2, 2002, pp. 61–75, https://doi.org/10.1109/42.993126. Accessed 16 Dec. 2022.